

TINGXI LI

800 W Campbell Rd, Richardson, TX 75080

☎ (213)-795-5275

✉ tingxi.li@utdallas.edu

🌐 [linkedin](#)

🏠 [homepage](#)

🎓 [Google Scholar](#)

Research Interests

I am broadly interested in the **inference-time efficiency** of deep learning systems across two levels: (i) kernel and subgraph-level efficiency, and (ii) model-level dynamicity. My primary research interests lie in **deep learning compilation**, targeting compiler infrastructure and AI-driven kernel generation. I also investigate dynamic behavior vulnerabilities in deep learning models and develop defenses against efficiency exploitation.

Publications

Characterizing Real-World Bugs in Tile Programs for Automated Bug Detection

R. Rathnasuriya, Z. Song, N. Majoju, **T. Li**, A. Moharir, W. Yang, T. Xie

ISSTA'26

SoK: Efficiency Robustness of Dynamic Deep Learning Systems

R. Rathnasuriya, **T. Li**, Z. Xu, Z. Song, M. Haque, S. Chen, W. Yang

USENIX Sec'25

COMET: Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models

Z. Xu, **T. Li**, R. Rathnasuriya, Z. Song, J. Ren, B. Mandalapu, S. Setayeshpour, X. Du, W. Yang

Technical Report

Practical: Retargeting AI Kernel DSLs Beyond GPUs: An Experience Report on Refactoring

TileLang to Sophgo TPUs

T. Ren, **T. Li**, X. Xiang, C. Xu, W. Yang, T. Xie

Under Review

An Empirical Study of GPU Kernel Performance Gaps in Modern Domain-Specific Languages

T. Li, R. Rathnasuriya, W. Yang

Under Review

AESOP: Adversarial Execution-path Selection to Overload Deep Learning Pipelines

T. Li, M. Ji, R. Rathnasuriya, S. Chen, Y. Hu, W. Yang

arXiv Preprint

Research Projects

A Survey of Automation in Kernel Generalization and Optimization

Jan. 2026 – Present

- Systematized DL compilation research into AI-driven kernel generation and compiler infrastructure, with subcategories on search space pruning, auto-tuning, cost modeling, and candidate validation.
- Analyzed IR evolution from loop-based to tile-based abstractions and its downstream impact on the compilation ecosystem.

Amazon Trusted AI Competition

Nov. 2024 – Jul. 2025

- Finalist team (\$250K prize) targeting black-box jailbreaking of CodeLLMs.
- Fine-tuned surrogate models for adversarial prompt guidance, achieving 0.80+ recall and 0.75+ F1 in malicious intent detection.

Education

University of Texas at Dallas

Sep. 2024 – Present

Doctor of Philosophy

Dalian University of Technology

Sep. 2019 – May 2024

Bachelor of Science

Internship Experience

Sophgo	Jun. 2024 – Aug. 2024
<i>Research Intern</i>	<i>Shenzhen, China</i>
• Developed C++ API code for models on RISC-V processors and authored deployment documentation. • Fine-tuned models on private datasets, identified failure cases, and applied data augmentation to improve robustness.	

Teaching

CS 4365 – Artificial Intelligence	Spring 2026
<i>Teaching Assistant, UT Dallas</i>	
CS 4375 – Introduction to Machine Learning	Fall 2025; Fall 2024
<i>Teaching Assistant, UT Dallas</i>	
CS 4337 – Programming Language Paradigms	Spring 2025
<i>Teaching Assistant, UT Dallas</i>	

Miscellaneous

Tech Stack: Python; C; C++; Java; PyTorch; L^AT_EX; SQL